Employee Performance Mapping

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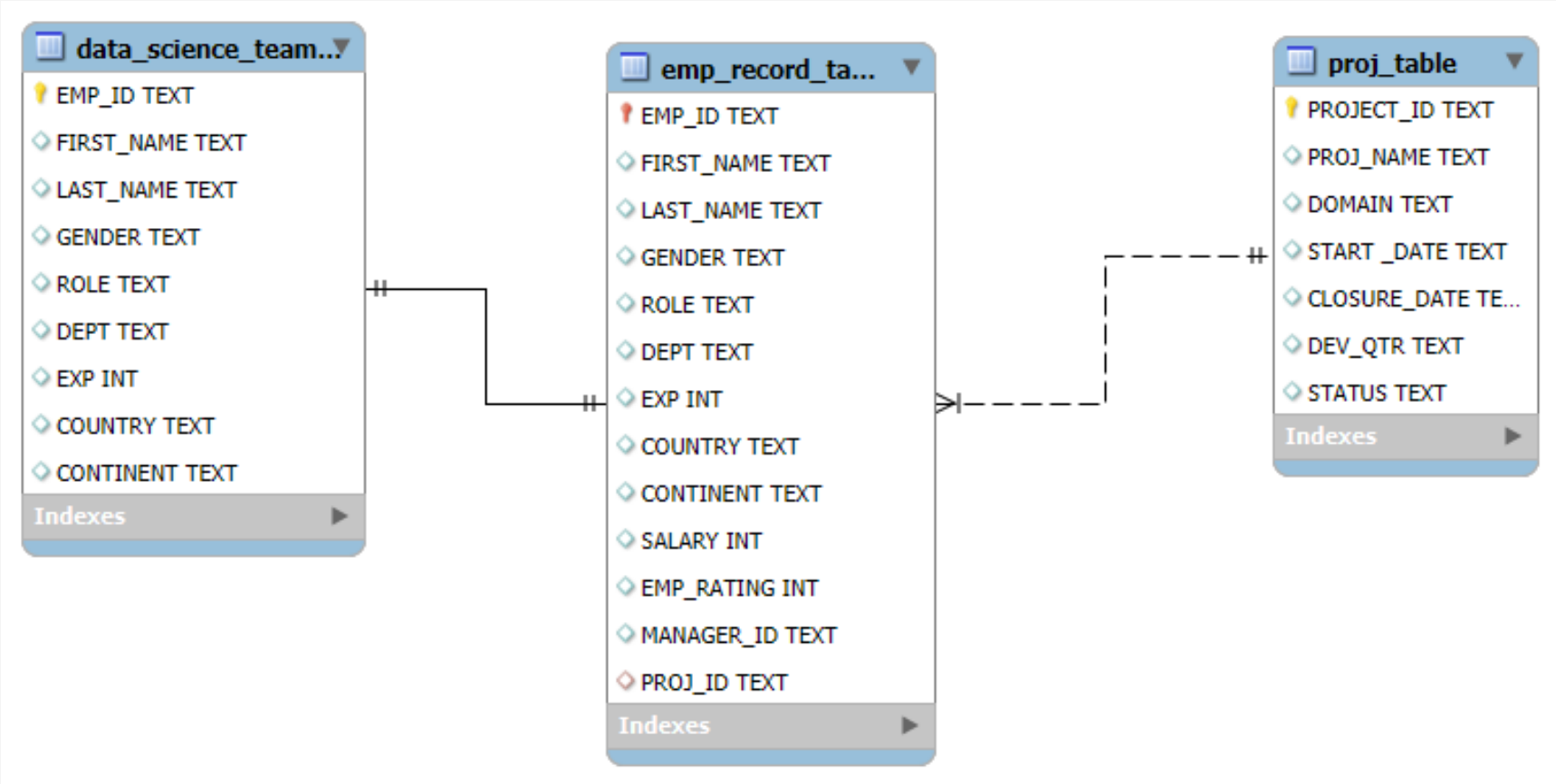
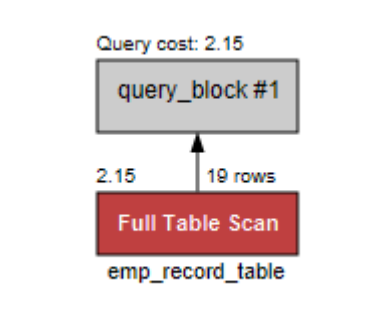
1. Create an ER diagram for the given employee database.



2. Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPARTMENT from the employee record table, and make a list of employees and details of their department.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT
E001	Arthur	Black	M	ALL
E005	Eric	Hoffman	M	FINANCE
E010	William	Butler	M	AUTOMOTIVE
E052	Dianna	Wilson	F	HEALTHCARE
E057	Dorothy	Wilson	F	HEALTHCARE
E083	Patrick	Voltz	M	HEALTHCARE
E103	Emily	Grove	F	FINANCE
E204	Karene	Nowak	F	AUTOMOTIVE
E245	Nian	Zhen	M	RETAIL
E260	Roy	Collins	M	RETAIL
E403	Steve	Hoffman	M	FINANCE
E428	Pete	Allen	M	AUTOMOTIVE
E478	David	Smith	M	RETAIL
E505	Chad	Wilson	M	HEALTHCARE
E532	Claire	Brennan	F	AUTOMOTIVE
E583	Janet	Hale	F	RETAIL
E612	Tracy	Norris	F	RETAIL
E620	Katrina	Allen	F	RETAIL
E640	Jenifer	Jhones	F	RETAIL



3. Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPARTMENT, and EMP_RATING if the EMP_RATING is:
less than two
greater than four
between two and four

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT	EMP_RATING
E001	Arthur	Black	M	ALL	5
E005	Eric	Hoffman	M	FINANCE	3
E010	William	Butler	M	AUTOMOTIVE	2
E052	Dianna	Wilson	F	HEALTHCARE	5
E057	Dorothy	Wilson	F	HEALTHCARE	1
E083	Patrick	Voltz	M	HEALTHCARE	5
E103	Emily	Grove	F	FINANCE	4
E204	Karene	Nowak	F	AUTOMOTIVE	5
E245	Nian	Zhen	M	RETAIL	2
E260	Roy	Collins	M	RETAIL	3
E403	Steve	Hoffman	M	FINANCE	3
E428	Pete	Allen	M	AUTOMOTIVE	4
E478	David	Smith	M	RETAIL	4
E505	Chad	Wilson	M	HEALTHCARE	2
E532	Claire	Brennan	F	AUTOMOTIVE	1
E583	Janet	Hale	F	RETAIL	2
E612	Tracy	Norris	F	RETAIL	4
E620	Katrina	Allen	F	RETAIL	1
E640	Jenifer	Jhones	F	RETAIL	4

4. Write a query to concatenate the FIRST_NAME and the LAST_NAME of employees in the *Finance* department from the employee table and then give the resultant column alias as NAME.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

NAME
Eric Hoffman
Emily Grove
Steve Hoffman

5. Write a SQL query to retrieve the employee ID, first name, role, and department of employees who hold leadership positions (Manager, President, or CEO).

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	ROLE	DEPT
E001	Arthur	PRESIDENT	ALL
E083	Patrick	MANAGER	HEALTHCARE
E103	Emily	MANAGER	FINANCE
E428	Pete	MANAGER	AUTOMOTIVE
E583	Janet	MANAGER	RETAIL
E612	Tracy	MANAGER	RETAIL

6. Write a query to list all the employees from the healthcare and finance departments using the union. Take data from the employee record table.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

NAME	DEPT
Chad Wilson	HEALTHCARE
Dianna Wilson	HEALTHCARE
Dorothy Wilson	HEALTHCARE
Emily Grove	FINANCE
Eric Hoffman	FINANCE
Patrick Voltz	HEALTHCARE
Steve Hoffman	FINANCE

7. Write a query to list employee details such as EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPARTMENT, and EMP_RATING grouped by dept. Also include the respective employee rating along with the max emp rating for the department.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	MAX_EMP_RATING_FOR_DEPT	EMP_RATING
E001	Arthur	Black	PRESIDENT	ALL	5	5
E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	12	2
E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	12	5
E428	Pete	Allen	MANAGER	AUTOMOTIVE	12	4
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	12	1
E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	10	3
E103	Emily	Grove	MANAGER	FINANCE	10	4
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	FINANCE	10	3
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	13	5
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	13	1
E083	Patrick	Voltz	MANAGER	HEALTHCARE	13	5
E505	Chad	Wilson	ASSOCIATE DATA SCIENTIST	HEALTHCARE	13	2
E245	Nian	Zhen	SENIOR DATA SCIENTIST	RETAIL	20	2
E260	Roy	Collins	SENIOR DATA SCIENTIST	RETAIL	20	3
E478	David	Smith	ASSOCIATE DATA SCIENTIST	RETAIL	20	4
E583	Janet	Hale	MANAGER	RETAIL	20	2
E612	Tracy	Norris	MANAGER	RETAIL	20	4
E620	Katrina	Allen	JUNIOR DATA SCIENTIST	RETAIL	20	1
E640	Jenifer	Jhones	JUNIOR DATA SCIENTIST	RETAIL	20	4

8. Write a query to calculate the minimum and the maximum salary of the employees in each role. Take data from the employee record table.

Using Windows Function

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FIRST_NAME	ROLE	MIN_SALARY	MAX_SALARY	SALARY
Steve	ASSOCIATE DATA SCIENTIST	4000	5000	5000
David	ASSOCIATE DATA SCIENTIST	4000	5000	4000
Chad	ASSOCIATE DATA SCIENTIST	4000	5000	5000
Claire	ASSOCIATE DATA SCIENTIST	4000	5000	4300
Katrina	JUNIOR DATA SCIENTIST	2800	3000	3000
Jenifer	JUNIOR DATA SCIENTIST	2800	3000	2800
Eric	LEAD DATA SCIENTIST	8500	9000	8500
William	LEAD DATA SCIENTIST	8500	9000	9000
Patrick	MANAGER	8500	11000	9500
Emily	MANAGER	8500	11000	10500
Pete	MANAGER	8500	11000	11000
Janet	MANAGER	8500	11000	10000
Tracy	MANAGER	8500	11000	8500
Arthur	PRESIDENT	16500	16500	16500
Dianna	SENIOR DATA SCIENTIST	5500	7700	5500
Dorothy	SENIOR DATA SCIENTIST	5500	7700	7700
Karene	SENIOR DATA SCIENTIST	5500	7700	7500
Nian	SENIOR DATA SCIENTIST	5500	7700	6500
Roy	SENIOR DATA SCIENTIST	5500	7700	7000

Using Group by Clause

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ROLE	MIN_SALARY	MAX_SALARY
PRESIDENT	16500	16500
LEAD DATA SCIENTIST	8500	9000
SENIOR DATA SCIENTIST	5500	7700
MANAGER	8500	11000
ASSOCIATE DATA SCIENTIST	4000	5000
JUNIOR DATA SCIENTIST	2800	3000

9. Write a query to assign ranks to each employee based on their experience. Take data from the employee record table.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID	rank
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL	NULL	1
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL	2
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL	3
E428	Pete	Allen	M	MANAGER	AUTOMOTIVE	14	GERMANY	EUROPE	11000	4	E001	NULL	3
E583	Janet	Hale	F	MANAGER	RETAIL	14	COLOMBIA	SOUTH AMERICA	10000	2	E001	NULL	3
E612	Tracy	Norris	F	MANAGER	RETAIL	13	INDIA	ASIA	8500	4	E001	NULL	4
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428	P204	5
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105	6
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083	P302	7
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	7500	5	E428	P204	8
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA	7000	3	E583	NA	9
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5	E083	P103	10
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	6500	2	E583	P109	10
E505	Chad	Wilson	M	ASSOCIATE DATA SCIEN...	HEALTHCARE	5	CANADA	NORTH AMERICA	5000	2	E083	P103	11
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIEN...	FINANCE	4	USA	NORTH AMERICA	5000	3	E103	P105	12
E478	David	Smith	M	ASSOCIATE DATA SCIEN...	RETAIL	3	COLOMBIA	SOUTH AMERICA	4000	4	E583	P109	13
E532	Claire	Brennan	F	ASSOCIATE DATA SCIEN...	AUTOMOTIVE	3	GERMANY	EUROPE	4300	1	E428	P204	13
E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA	3000	1	E612	P406	14
E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA	2800	4	E612	P406	15

10. Write a query to create a view that displays employees in various countries whose salary is more than six thousand. Take data from the employee record table.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Name	COUNTRY	SALARY
Arthur Black	USA	16500
Eric Hoffman	USA	8500
William Butler	FRANCE	9000
Dorothy Wilson	USA	7700
Patrick Voltz	USA	9500
Emily Grove	CANADA	10500
Karene Nowak	GERMANY	7500
Nian Zhen	CHINA	6500
Roy Collins	INDIA	7000
Pete Allen	GERMANY	11000
Janet Hale	COLOMBIA	10000
Tracy Norris	INDIA	8500

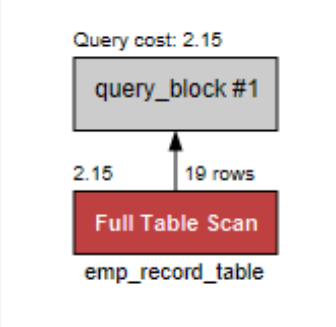
11. Write a nested query to find employees with experience of more than ten years. Take data from the employee record table.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

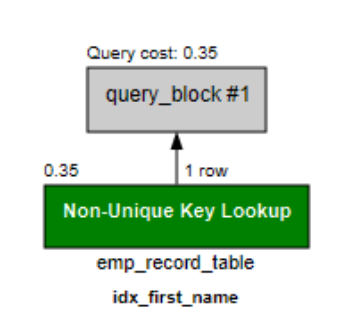
Name	EXP
Arthur Black	20
Eric Hoffman	11
William Butler	12
Patrick Voltz	15
Emily Grove	14
Pete Allen	14
Janet Hale	14
Tracy Norris	13

12. Create an index to improve the cost and performance of the query to find the employee whose FIRST_NAME is 'Eric' in the employee table after checking the execution plan.

Before creating Index



After creating Index



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	emp_record_table	NULL	ref	idx_first_name	idx_first_name	203	const	1	100.00	Using where

13. Write a query to calculate the bonus for all the employees, based on their ratings and salaries (Use the formula: 5% of salary * employee rating).

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	Name	ROLE	SALARY	EMP_RATING	Bonus
E001	Arthur Black	PRESIDENT	16500	5	4125.00
E005	Eric Hoffman	LEAD DATA SCIENTIST	8500	3	1275.00
E010	William Butler	LEAD DATA SCIENTIST	9000	2	900.00
E052	Dianna Wilson	SENIOR DATA SCIENTIST	5500	5	1375.00
E057	Dorothy Wilson	SENIOR DATA SCIENTIST	7700	1	385.00
E083	Patrick Voltz	MANAGER	9500	5	2375.00
E103	Emily Grove	MANAGER	10500	4	2100.00
E204	Karene Nowak	SENIOR DATA SCIENTIST	7500	5	1875.00
E245	Nian Zhen	SENIOR DATA SCIENTIST	6500	2	650.00
E260	Roy Collins	SENIOR DATA SCIENTIST	7000	3	1050.00
E403	Steve Hoffman	ASSOCIATE DATA SCIEN...	5000	3	750.00
E428	Pete Allen	MANAGER	11000	4	2200.00
E478	David Smith	ASSOCIATE DATA SCIEN...	4000	4	800.00
E505	Chad Wilson	ASSOCIATE DATA SCIEN...	5000	2	500.00
E532	Claire Brennan	ASSOCIATE DATA SCIEN...	4300	1	215.00
E583	Janet Hale	MANAGER	10000	2	1000.00
E612	Tracy Norris	MANAGER	8500	4	1700.00
E620	Katrina Allen	JUNIOR DATA SCIENTIST	3000	1	150.00
E640	Jenifer Jhones	JUNIOR DATA SCIENTIST	2800	4	560.00

14. Write a query to calculate the average salary distribution based on the continent and country. Take data from the employee record table.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CONTINENT	COUNTRY	avg(SALARY)
ASIA	CHINA	6500.0000
ASIA	INDIA	6166.6667
EUROPE	FRANCE	9000.0000
EUROPE	GERMANY	7600.0000
NORTH AMERICA	CANADA	7000.0000
NORTH AMERICA	USA	9440.0000
SOUTH AMERICA	COLOMBIA	5600.0000

15. Add CHECK constraint on EMP_RATING to allow only values between 1 and 5. Add UNIQUE constraint on combination of FIRST_NAME and LAST_NAME.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EMP_ID	text	YES		NULL	
FIRST_NAME	text	YES	MUL	NULL	
LAST_NAME	text	YES		NULL	
GENDER	text	YES		NULL	
ROLE	text	YES		NULL	
DEPT	text	YES		NULL	
EXP	int	YES		NULL	
COUNTRY	text	YES		NULL	
CONTINENT	text	YES		NULL	
SALARY	int	YES		NULL	
EMP_RATING	int	YES		NULL	
MANAGER_ID	text	YES		NULL	
PROJ_ID	text	YES		NULL	

16. Write a query that demonstrates the use of string, numeric, and date/time functions by performing the following: a. Extract the first three letters of each employee's FIRST_NAME from the emp_record_table
b. Calculate the square of the EMP_RATING using a numeric function
c. Extract the month from the START_DATE in the proj_table

a.

Result Grid

First_3_Letters
Art
Eri
Wil
Dia
Dor
Pat
Emi
Kar
Nia
Roy
Ste
Pet
Dav
Cha
Cla
Jan
Tra
Kat
Jen

b.

Result Grid Filter Rows:

EMP_ID	EMP_RATING	Rating_Squared
E001	5	25
E005	3	9
E010	2	4
E052	5	25
E057	1	1
E083	5	25
E103	4	16
E204	5	25
E245	2	4
E260	3	9
E403	3	9
E428	4	16
E478	4	16
E505	2	4
E532	1	1
E583	2	4
E612	4	16
E620	1	1
E640	4	16

c. I think there is some mistake in the table's column name.

Result Grid Filter Rows:

START _DATE	Start_Month
04-06-2021	NULL
04-11-2021	NULL
04-12-2021	NULL
07/15/2021	NULL
10-08-2021	NULL
07-09-2021	NULL

17. Write two queries using subqueries to modify the dataset:
a. Set EMP_RATING = 1 for employees with experience less than 1 year
b. Remove employees from emp_record_table whose salary is below 2000 and who are not assigned to any project.

a. ans: There are no employees who have less than 1 year of experience.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL	NULL
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428	P204
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5	E083	P103
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083	P302
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	7500	5	E428	P204
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	6500	2	E583	P109
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA	7000	3	E583	NA
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIEN...	FINANCE	4	USA	NORTH AMERICA	5000	3	E103	P105
E428	Pete	Allen	M	MANAGER	AUTOMOTIVE	14	GERMANY	EUROPE	11000	4	E001	NULL
E478	David	Smith	M	ASSOCIATE DATA SCIEN...	RETAIL	3	COLOMBIA	SOUTH AMERICA	4000	4	E583	P109
E505	Chad	Wilson	M	ASSOCIATE DATA SCIEN...	HEALTHCARE	5	CANADA	NORTH AMERICA	5000	2	E083	P103
E532	Claire	Brennan	F	ASSOCIATE DATA SCIEN...	AUTOMOTIVE	3	GERMANY	EUROPE	4300	1	E428	P204
E583	Janet	Hale	F	MANAGER	RETAIL	14	COLOMBIA	SOUTH AMERICA	10000	2	E001	NULL
E612	Tracy	Norris	F	MANAGER	RETAIL	13	INDIA	ASIA	8500	4	E001	NULL
E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA	3000	1	E612	P406
E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA	2800	4	E612	P406

17. Write two queries using subqueries to modify the dataset:
a. Set EMP_RATING = 1 for employees with experience less than 1 year
b. Remove employees from emp_record_table whose salary is below 2000 and who are not assigned to any project.

b. ans: There are no employees whose salary is below 2000 and who are not assigned to any project.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL	NULL
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428	P204
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5	E083	P103
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083	P302
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	7500	5	E428	P204
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	6500	2	E583	P109
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA	7000	3	E583	NA
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIEN...	FINANCE	4	USA	NORTH AMERICA	5000	3	E103	P105
E428	Pete	Allen	M	MANAGER	AUTOMOTIVE	14	GERMANY	EUROPE	11000	4	E001	NULL
E478	David	Smith	M	ASSOCIATE DATA SCIEN...	RETAIL	3	COLOMBIA	SOUTH AMERICA	4000	4	E583	P109
E505	Chad	Wilson	M	ASSOCIATE DATA SCIEN...	HEALTHCARE	5	CANADA	NORTH AMERICA	5000	2	E083	P103
E532	Claire	Brennan	F	ASSOCIATE DATA SCIEN...	AUTOMOTIVE	3	GERMANY	EUROPE	4300	1	E428	P204
E583	Janet	Hale	F	MANAGER	RETAIL	14	COLOMBIA	SOUTH AMERICA	10000	2	E001	NULL
E612	Tracy	Norris	F	MANAGER	RETAIL	13	INDIA	ASIA	8500	4	E001	NULL
E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA	3000	1	E612	P406
E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA	2800	4	E612	P406